



Problems (Taylor Series - Univariate)

1. Write the Taylor series expansion for the following
 - a. $f(x) = e^{-x}$, at $x=0$
 - b. $f(x) = e^{2x}$, at $x=0$
 - c. $f(x) = \sqrt{2+x}$, at $x = 1$
 - d. $f(x) = \ln(2+9x)$, at $x = 1$
2. An exotic equity derivative has the following valuation function. $V = S \ln(S)$. Currently, the stock price is $S=\$100$. Approximate the change in value of the derivative if the stock price appreciates by \$1. (Hint: Use Taylor series expansion till 2nd order)